STIDK3013 RESEARCH METHOD IN IT

Session 2024/2025 (A241)

Assignment 1: Research Proposal

Title: The Impact of IoT-enabled Smart Classroom for University Students Quality Performance

SUBMITTED BY:

NO	NAME	MATRIC NO.
1.	NURUL HARISYAH BINTI ADZAMI	293174
2.	NUR SYAZWANI AMNI BINTI MOHD REDZUAN	292376
3	NURUL AZALIA BINTI RAZAK	292743
4	AZIEATUL AZIELA BINTI ZULKEFLI	292845

SUBMITTED TO:

DR. AMIRULIKHSAN ZOLKAFLI

1.0 INTRODUCTION

a. Research Background

IoT-enabled smart classrooms are transforming university institutions that combine advanced technologies that create interactive and data-driven learning environments. Smart classrooms equipped with IoT tools such as smart boards, real-time data analytic, automated attendance, environmental sensors and wifi offer university students a fresh approach to engage with the academic materials. These environments also support effective resource management and personalized learning.

b. Problem Statement

Studying how IoT-enabled smart classrooms technologies affect students' performance in learning experiences, engagement levels, and academic achievement is the main goal of this study. The study intends to give universities and educators ideas on how to successfully utilize smart classrooms to improve educational results by examining the benefits and difficulties related to its adoption in educational contexts. Furthermore, this study aims to investigate how smart classrooms might enhance academic achievement, learning experience, and raise the standard of students' classroom experiences.

c. Research Questions

1. Which specific smart classroom features are most effective in enhancing the university student's learning experience?
2. How does the quality of smart classrooms' features affect university students's performance and effectiveness?
3. To what extent do smart classroom features influence university student academic performance and engagement?

d. Research Objectives

1. To identify which IoT-enabled smart classroom features that enhance learning experiences at the university level for university students.

2. To assess the effectiveness and performance of smart classrooms based on their quality.

3. To evaluate the extent to which smart classroom features impact student academic performance and engagement.

e. Research Scope

- Analysis of how IoT tools improve students' learning and their performance.
- Analysis of IoT technology's role in improving student's grades and academic performance.
- Identify which IoT features are most effective in enhancing students academic performance.
- Investigate how students view and their acceptance towards IoT-enabled smart classrooms.

f. Research Significance/Implication

- This research shows how smart classrooms with IoT features can improve students' academic performance and engagement.
- This research also will show how an IoT-enabled smart classroom can create an interactive learning environment.
- The finding can guide universities to use IoT learning technology effectively to improve the quality of education.
- The findings can also inspire further studies on using technology in education.

2.0 LITERATURE REVIEW

Literature Review The Impact of IoT-enabled Smart Classroom for University Students Quality Performance

The Internet of Things (IoT) has revolutionized education, particularly in the smart classroom where it facilitates flexible and dynamic learning environments for college students. Hence, to improve the experience of learning, institutions of education have implemented smart classrooms to provide cutting-edge solutions such as interactive boards, automatic attendance and real-time analytics.

1. **Student Intention to Use IoT Services in Smart Classrooms**
   IoT services in smart classrooms desired to be utilized by university students' are examined in research by Alhasan et al. (2023). According to it, using tools like interactive whiteboards provide active learning and helps students to interact with subjects in more conventional settings.

2. **Infrastructure and Readiness**
   The problems of IoT adoption in higher education was examined in the study by Negm (2022), with a particular emphasis on how to prepare institutions to use the technologies. Eventually, IoT in smart classrooms requires a robust technology infrastructure, which involves fast internet, well-maintained and data security.

3. **Enhanced Interaction and Real-World Application**
   Interactive resources have been offered. For instance, digital whiteboards, sensors, and smart gadgets that create the learning environment more adaptable and responsive, the usage of IoT in classrooms provide hands-on learning (Kaur et al., 2022). In addition, IoT technology can help close the gap between theoretical knowledge and practical abilities by enabling students to engage with industry standard tools in an academic context (Zainuddin et al., 2021).

4. **Environmental and Sustainability Considerations**
   According to Rukhiran et al. (2022), IoT-based technologies also support sustainability in educational institutions. Having IoT it can help smart schools adopt sustainable practices like eco-friendly gadgets and have enough energy classrooms. Smart classrooms provide global sustainability goals by reducing waste and encouraging environmental responsibility throughout automated technologies.

5. **Challenges of IoT Adoption in Education**
   However, there are disadvantages in applied IoT in education, such as data privacy, high expenses and the requirement of ongoing technical assistance. It is because IoT devices keep and store sensitive data of students. Hence, the university itself must have strong cybersecurity procedures. Negm (2022).

In conclusion, IoT in smart classrooms offers students better learning chances, to create an educational environment that's more accessible and eco-friendly but institutions require to overcome security issues.

3.0 PROPOSED METHODOLOGY

This research will use a mixed-methods strategy, integrating quantitative and qualitative data. This method allows us to gain a more comprehensive understanding of the effects of IoT-enabled smart classrooms on university students' engagement, performance, and attitudes.

1. Population and sample
    - University students who use IoT-enabled smart classrooms.
    - A group of about 250 students for surveys was selected to get a variety of perspectives.
2. Data collection methods
    - Quantitative Data Collection
        - Survey: Students will be ask to complete an online survey about their engagement, academic performance, and satisfaction with IoT features in smart classrooms.
        - Classroom data: Permission will be obtained to use classroom IoT system data to evaluate student engagement.
    - Qualitative Data Collection
        - Interviews: We will interview students to learn about their experiences, challenges, and thoughts on smart classroom technology.
3. Data Analysis Techniques
    - Quantitative Analysis
        - Descriptive statistics: Analyze general trends in survey responses.
    - Qualitative Analysis
        - Sentiment Check: Assess overall student attitudes toward smart classrooms.
4. Tools and instruments: Survey, Interview, IoT Classroom Data
5. Ethical Considerations
    - Consent: Participants will be made aware of the study's objectives and have the option to decide if they want to take part or not.
    - Data Privacy: Data will be anonymized and stored securely.

- Confidentiality: Only aggregated data will be reported.

6. Study Limitations
    - Generalizability: The results may only be relevant to the particular universities examined.
    - Tech Differences: The IoT configurations vary among universities, potentially impacting outcomes.
    - Response bias: Students may report in ways that are different from how they truly feel or act.

4.0 EXPECTED RESULTS/FINDINGS

The expected findings of this research indicate that by enhancing learning through personalisation and interaction, IoT-enabled smart classrooms will boost academic performance and student engagement. The learning process will be improved by important features like smart boards, automated attendance, and smart projectors, but there may be disadvantages like technical difficulties, privacy issues, and expensive prices. Students may have a favorable opinion of IoT tools but could raise issues about data privacy and the importance of technical assistance. The study will give universities actionable recommendations for leveraging IoT to enhance educational results, deliver more tailored experiences, and maximize resource efficiency.

REFERENCES

Ameer Alhasan, Hussein, M. H., Lukman Audah, Ammar Al-Sharaa, Ibrahim, I., & Mahmoud, M. A. (2023). *A case study to examine undergraduate students' intention to use internet of things (IoT) services in the smart classroom*. https://doi.org/10.1007/s10639-022-11537-z

Kaur, A., Bhatia, M., & Stea, G. (2022). A Survey of Smart Classroom Literature. *Education Sciences*, *12*(2), 86. https://doi.org/10.3390/educsci12020086

Negm, E. (2022). Intention to use Internet of Things (IoT) in higher education online learning – the effect of technology readiness. *Higher Education, Skills and Work-Based Learning*, *13*(1). https://doi.org/10.1108/heswbl-05-2022-0121

Rukhiran, M., Phaokla, N., & Netinant, P. (2022). Adoption of Environmental Information Chatbot Services Based on the Internet of Educational Things in Smart Schools: Structural Equation Modeling Approach. *Sustainability*, *14*(23), 15621. https://doi.org/10.3390/su142315621

Zainuddin, A. A., Bhattacharjee, S., Kalliat, S., Shrestha, S., Sivaraman, S., Khalique, M. M., Naidu, D. R., & Manokaran, P. (2021). Trends and Challenges of Internet-of-Things in the Educational Domain. *Malaysian Journal of Science and Advanced Technology*, 81–88. https://doi.org/10.56532/mjsat.v1i3.17